Qianxun (Cecilia) Xu

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EDUCATION

Duke Kunshan University (DKU) & Duke University Dual Degree 2022 - 2026

- B.S. in Applied Math and Computer Science; Computer Science Track (GPA: 3.922/4.0)
- Courses: Introduction to Machine Learning, Design and Analysis of Algorithms, Stochastic Processes, Artificial Intelligence.
- Scholarships: Merit-based College Entrance Scholarship covering 50% tuition.

ACADEMIC PAPERS

Xu, Q., Li, Z. Partial Physics Informed Diffusion Model for Ocean Chlorophyll Reconstruction. *NeurIPS* 2025 main track (poster), accepted.

Xu, Q., Boukerche, A., Sun, P. FENSe: Feedback-Enabled Neighbor Selection for Spatial Aware Collaborative Perception. *IEEE International Conference on Parallel and Distributed Systems, ICPADS 2025*, accepted.

Xu, Q., Song, C., Cai, Y., Zhang, C. Temporal-aware Multi-event Video Generation. Submitted to *CVPR* 2026, under review.

RESEARCH EXPERIENCE

Partial Physics Informed Diffusion Model (PPIDM), ***DKU Summer Research Scholar Program***

Lead Researcher May 2024 - May 2025

Supervisor: prof. Zuchuan Li, DKU Kunshan, China

- Built PPIDM to reconstruct real world states from sparse data under uncertain physics. Integrated incomplete physics knowledge into the diffusion process via a novel physics residual difference loss that selectively encodes measurable processes while avoiding over-constraints on unknown processes.
- Designed differentiable physics operators and a hyperparameter tuning guide for different dominance states of known physics.
- Evaluated PPIDM on long-range infilling and forecasting of Southern Ocean chlorophyll with physics-aware metrics. Outperformed data-only diffusion baselines by 34% lower reconstruction errors with improved stability, and baselines that misapply incomplete physics by 49%.

Feedback-Enabled Neighbor Selection (FENSe), ***DKU Undergraduate Signature Work***

Lead Researcher Jan - Aug 2025

Supervisor: prof. Peng Sun, DKU Kunshan, China

- Addressed communication redundancy among neighbor vehicles via a plug and play selection module that converts single vehicle outputs into compact feedback signatures, performs dynamic clustering and ranking of collaborators, and prioritizes transmission of maximally informative features to the ego vehicle.
- Performed thorough ablations and evaluations, achieving a stronger accuracy-efficiency trade-off (+4% accuracy at equal bandwidth) with lower ego side compute.

AIGC Summer Research, ***Westlake University***

Lead Researcher May - Nov 2025

Supervisor: prof. Chi Zhang, Westlake University Hangzhou, China

- **Temporal-aware training-free multi-event video generation**. Addresses time-agnostic cross attention in video diffusion model via (i) a novel projection-based guidance framework that aligns frames with intended events, and (ii) an adaptive solver that balances guidance strength to generate coherent multi-event videos.
- **Region-constrained 2D personalization & multi-concept learning**: Integrated mask guidance to LoRA-style fine tuning through refined attention masks or YOLO+SAM derived masks to prevent background leakage; enabled simultaneous multi-concept extraction to reduce concept blending. (GitHub link)
- **Scale and semantics aware video object replacement**: Extended inversion-free video editing pipeline with a late step fixed noise schedule, smoothing editing trajectories and reducing flicker in newly revealed regions while supporting transformations with large geometric and semantic shifts between source and target objects. (GitHub link)

PROJECT EXPERIENCE

TransGCN For Muscle Activation Prediction from Joint Angles During Motion Mar - May 2025

Individual Course Project Kunshan, China

- Adapted a spatio-temporal Transformer together with a kinematic graph convolutional network to predict frame wise muscle activations from joint angle sequences. Integrated demographic embeddings (gender/weight/height) for personalized estimates.
- Performed systematic ablations, achieving >11% improvement over various LSTM and vanilla Transformer baselines.
- Delivered 3 milestone reports, an 8-page final report, and a 90-minute poster presentation. (GitHub link)

CAMPUS EXPERIENCE

Computer Science 101 Course, ***DKU*** Oct - Dec 2022

Teaching Assistant Kunshan, China

- Led weekly lab sessions and office hours; graded quizzes and assignments; analyzed error trends and common confusions.
- Utilized data-driven insights to recommend curriculum adjustments, increasing student engagement and academic achievement.

ADDITIONAL INFORMATION

Languages: English (GRE:324/340+4.5/6; TOEFL:116/120), German (intermediate), Mandarin (native).

Skills: Python, PyTorch, Java, C, Latex, Excel, Adobe PS.